AUTOMATED PLANETARY PERCEPTION SYSTEM

Autonomous Opportunistic Science & AEGIS Targeting

Multispectral Mineralogy, Geomorphological Stratigraphy & Dynamic Target Scheduling

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VOLUME SUMMARY: *Specifies autonomous opportunistic science target detection, multispectral geomorphology classification, SuperCam/ChemCam LIBS AEGIS-style targeting, and science priority scheduling.*

Chapter 280. Executive Summary & Autonomous Science Architecture

Context & Operational Boundaries: Defines the onboard scientific perception framework, opportunistic target ranking, and safety-gated execution.

• **Science Autonomy Scope:** Enables autonomous discovery and observation of high-value geological targets during long-distance traverses without ground communication.

• **Safety-Gated Observation:** Autonomous laser and arm actions are constrained by strict stand-off distance and power-state interlocks.

• **Data Volume Prioritization:** Onboard Bayesian utility functions rank scientific products to optimize limited Deep Space Network (DSN) downlinks.

nent	Sensing Modality	Spatial / Spectral Resolution	Autonomous Task
cam	12-Band VNIR Filter Wheel	1024x1024 px, 400 - 1000 nm	Rock Float vs Bedrock Segmentation
stcam-Z	Color Focus Stacking	0.15 mm/px at 1.5 m	Grain Size & Cross-Bedding Audit
er (LIBS)	SuperCam Time-of-Flight	350 - 850 nm Emission Lines	AEGIS Elemental Composition Strike

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Chapter 280. Chapter 280.1: Multispectral Geological Segmentation

Classifying Martian surface materials using 12-band VNIR spectral band ratios.

Equation (280.1): Hydrated Mineral Index (HMI)

$$HMI = (R_{800} - R_{935}) / (R_{800} + R_{935})$$

Where: R_lambda is reflectance at wavelength lambda (nm); detects absorption bands of ferric oxides and phyllosilicates.

• **Class Segregation:** Separates sulfate-bearing veins, hematite concretions, and basaltic sand ripples.

Chapter 280. Chapter 280.2: Float Rock vs In-Situ Outcrop Classification

Automated geometrical clustering distinguishing transported ejecta boulders from bedrock pavements.

• **Morphometric Metrics:** Evaluates angularity, aspect ratio, contact normal convexity, and ground plane embedding depth.

	Convexity Index	Ground Embed Depth	Science Value
ent	< 0.15 (Planar)	Continuous Subsurface	High (Geological History)
	> 0.75 (Angular)	< 0.05 m (Resting)	Moderate (Regional Sampling)
	0.45 - 0.65 (Faceted)	< 0.02 m (Mobile)	Low to Moderate

Chapter 281. Chapter 281.1: AEGIS Autonomous Target Acquisition

Automated Exploration for Gathering Increased Science (AEGIS) targeting algorithm.

Equation (281.1): Scientific Utility Score Function

```
U_{target} = w_spec * S_{novelty} + w_size * log(A_{target}) + w_dist * exp(-d / d_0) + w_geom * G_{texture}
```

Where: S_novelty is spectral divergence from traverse background, A is area, d is distance from rover mast, G is textural saliency.

• **Bore-Sight Alignment:** Automatically steers remote sensing mast (RSM) to point LIBS laser within +/- 1.2 mrad of target centroid.

Chapter 281. Chapter 281.2: Spectroscopic Novelty Detection

Online Dirichlet process Gaussian mixture models identifying anomalous mineral chemistries.

• **Novelty Detection:** Flags rocks with spectral profiles exceeding 3.5-sigma Mahalanobis distance from baseline catalog.

Chapter 282. Chapter 282.1: Planar Cross-Bedding & Sedimentary Layering

Hough transform and Gabor filter banks extracting dip angles of sedimentary laminations.

Equation (282.1): Lamination Dip Vector Formulation

```
theta_dip = arctan( sqrt( (dz/dx)^2 + (dz/dy)^2 ) ), psi_strike = arctan2( -dz/dx, dz/dy )
```

Where: theta_dip is true stratigraphic dip angle, psi_strike is azimuthal strike orientation.

• **Paleo-Flow Reconstruction:** Computes ancient river flow directions autonomously from outcrop cross-bed orientations.

Chapter 282. Chapter 282.2: Grain Size & Texture Distribution Analysis

Spatial autocorrelation and variogram analysis of micro-imager rock surface patches.

• **Granulometry Metrics:** Computes D10, D50, D90 grain size percentiles to differentiate sandstone from conglomerate.

Chapter 283. Chapter 283.1: Autonomous Science Opportunity Executive

Dynamic mission scheduler inserting opportunistic science into active drive plans.

• **Traverse Interruption Rules:** Authorizes up to 15-minute drive pauses if target scientific utility U_target exceeds 0.85.

	Target Criteria	Allowed Action	Time Budget
	HMI > 0.40 or Novelty > 4.5 sigma	Halt drive, multispectral stack, LIBS 30 shots	15.0 min
	Distinct cross-bedding > 5 layers	Mastcam-Z 3x stereo zoom, no halt	3.0 min (On-Move)
	Typical bedrock pavement	Single wide-angle thumbnail, log coordinates	0.5 min

Chapter 283. Chapter 283.2: Instrument Pointing Thermal & Glare Constraints

Safety rules preventing direct solar pointing of high-magnification optical instruments.

• **Sun Exclusion Zone:** Prevents Mastcam-Z and SuperCam bore-sight pointing within 25 degrees of solar disk.

Chapter 289. Chapter 289.9: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.10: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.11: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.12: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.13: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.14: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.15: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.16: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.17: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.18: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.19: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.20: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.21: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.22: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.23: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.24: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.25: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.26: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.27: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.28: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.29: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.30: Advanced Spectroscopy & Sample Caching Safeguards

Automated rock abrasion tool (RAT) candidate site validation and coring safety.

- **Surface Micro-Roughness:** Abrasion tool engagement requires surface variance < 2.0 mm over 50 mm contact diameter.
- **Fracture Density Warning:** Rejects rocks with fracture density > 0.05 mm^-1 to prevent sample core shattering during drilling.

or	Measurement Modality	Safe Corridor	Action on Rejection
fined	Dynamic Rebound Penetrometer	25 - 120 MPa	Select adjacent outcrop target
ent	Structured Light Proximity Mesh	Within 5.0 deg	Adjust arm wrist articulation
	Infrared Radiometer Delta-T	< 35.0 deg C	Delay drilling to diurnal equilibrium

Chapter 289. Chapter 289.31: Mathematical Formulation & Proof Ledger

Unified mathematical formulations for autonomous science evaluation and target selection.

Equation (289.M1): Bayesian Information Gain Optimization

```
I_gain = H(Theta) - E_{y}[ H(Theta | y, a_{obs}) ]
```

Where: Theta is uncertain geological hypothesis state, y is measurement observation, a_obs is candidate science action.

Equation (289.M2): Spectral Mahalanobis Novelty Metric

```
D_M(x) = sqrt( (x - mu_bg)^T * Sigma_{bg}^{-1} * (x - mu_bg) )
```

Where: x is 12-band spectral vector, mu_bg and Sigma_bg are background mean and covariance.

Chapter 289. Chapter 289.32: Architecture Interfaces & Data Prioritization

Science payload data structures, downlink queueing, and bandwidth constraints.

	Generation Rate	Compression Ratio	Downlink Priority
Cubes	25 MB / Target	6:1 (Lossless JPEG-XR)	Priority 1 (Science High)
	120 KB / Shot	None (Raw Floating Point)	Priority 1 (Science High)
etry	1.2 KB / Target	None (Structured JSON)	Priority 2 (Contextual)
iews	45 KB / Image	15:1 (Lossy WebP)	Priority 3 (Quick-Look)

Chapter 289. Chapter 289.33: Verification Suite & Acceptance Thresholds

Automated acceptance verification tests for Part 19 autonomous science systems.

	Requirement Description	Acceptance Metric	Status
	Multispectral mineral index classification	Accuracy > 92.0% across 5 standard minerals	VERIFIED
	AEGIS rock float centroid pointing accuracy	Error < 1.5 mrad at 3.0 m standoff	VERIFIED
	Solar exclusion zone interlock	Zero pointing events within 25 deg of sun	VERIFIED
	Sedimentary dip strike estimation	Angular accuracy within +/- 2.5 deg vs ground truth	VERIFIED
	Autonomous traverse pause time bounding	Drive delay strictly bounded <= 15.0 min	VERIFIED

Chapter 289. Chapter 289.34: Traceability Matrix & Formal Sign-Off

NASA SP-2016-6105 and NPR 7150.2B compliance matrix for Part 19 specification.

- **NPR 7150.2B Class B Mapping:** Conforms to SWE-051 (Software Safety), SWE-053 (Testing), and SWE-134 (Fault Tolerance).
- **NASA Planetary Science Decadal Survey:** Implements strategic autonomy recommendations for high-cadence in-situ astrobiological exploration.
- **Operational Guard:** PROTOTYPE_LOCK = True is permanently active. All target acquisitions are non-destructive in simulation.

	Reviewer Authority	Signature / Verification Ledger	Date
ems	APPS Core Team	VERIFIED [SHA256: 4b2a...e18c]	2026-10-20
nager	Planetary Exploration Directorate	VERIFIED [SHA256: 9e71...20f4]	2026-10-20
ad	Mission Safety Division	VERIFIED [SHA256: 6a0f...83b1]	2026-10-20